# 2025-12-11 - Servidio (18)

Previous lecture: 2025-12-05 - Valentini (17)
Next lecture: 2025-12-12 - Servidio (19)

# Tearing instability (continuing)

Before we obtained the momentum equation

$$\gamma(v_y'' - k^2 v_y) = ik[B_{0x}(b_y'' - k^2 b_y) - b_y B_{0x}'']$$

For the magnetic field equation we have

$$\begin{aligned}
&\frac{\partial \bar{b}}{\partial t} = \nabla \times (\bar{v}_1 \times \bar{B}) + \eta \nabla^2 \bar{b} \\
\implies &\frac{\partial}{\partial t}\left[b_y(y)e^{\gamma t + ikx}\right] = -\partial_x\left[-v_y(y)B_{0x}e^{\gamma t + ikx}\right] + \eta(\partial_{xx}^2 + \partial_{yy}^2)b_y(y)e^{\gamma t + ikx} \\
\implies &\gamma b_y e^{\gamma t + ikx} = B_{0x}ikv_y e^{\gamma t + ikx} + \eta\left(-k^2 b_y + b_y''\right)e^{\gamma t + ikx} \\
\implies &\gamma b_y = B_{0x}ikv_y + \eta\left(-k^2 b_y + b_y''\right)
\end{aligned}$$

We rename the only two variables as

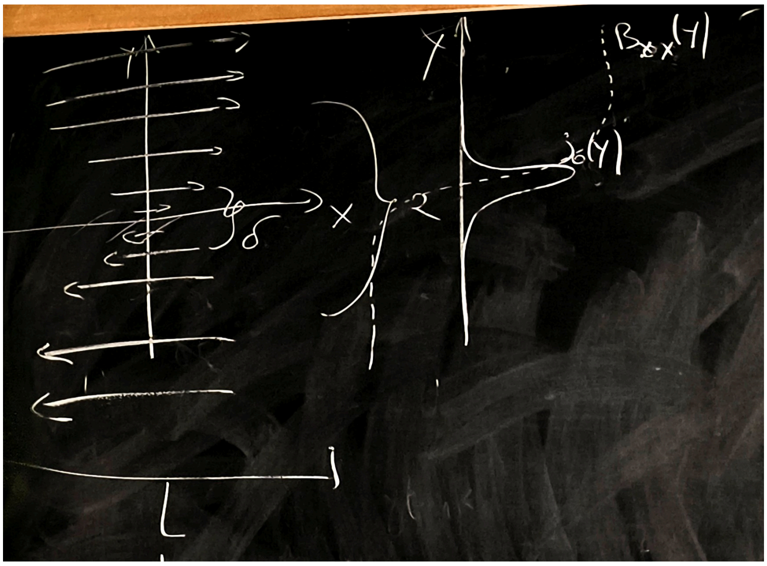
$$v_y \to v, \qquad b_y \to b$$

so

$$\begin{cases}
\gamma(v'' - k^2 v) = ik[B_{0x}(b'' - k^2 b) - b_y B_{0x}''] \\
\gamma b = B_{0x}ikv + \eta(-k^2 b + b'')
\end{cases}$$

with $B_{0x} = B_{0x}(y)$

PIC dot mag field (continuos line is j0)

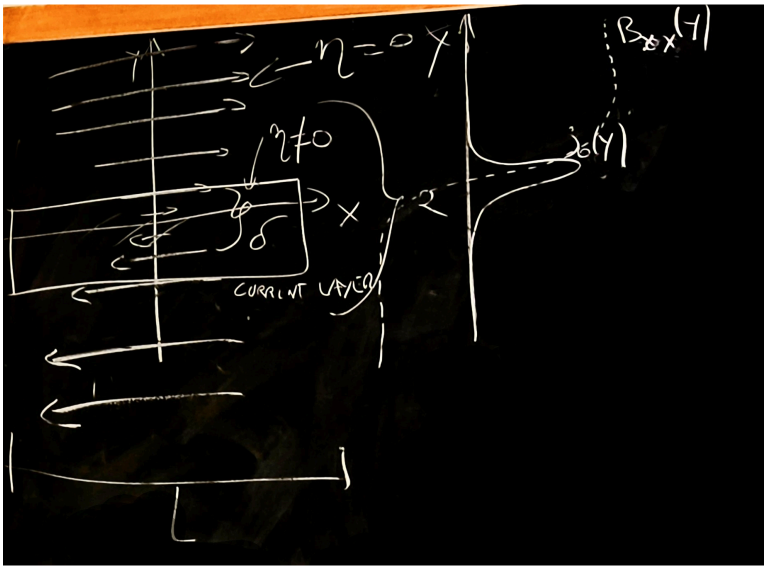



In the region $|y| \to +\infty$ we have $j_0 = 0$ and the maximum of $B_{0x}$ (outside)
In the region $|y| \to 0$ we have $B_{0x} \sim 0$ and the maximum of $j_0$ (inside the sheet)

$$L \gg a \gg \delta \qquad \Longrightarrow \qquad k = \frac{2\pi}{L} \ll \frac{1}{a} \ll \frac{1}{\delta}$$

we have a layer of current close to $y = 0$ and the current is only localized here

PIC same as before but with box




First we solve the system in the inside of the sheet, then we solve on the outside.
At the end we do the asymptotic matching.

# Inside

We have

$$\eta \neq 0, \qquad B_{0x} \sim 0 \ll v, b$$

We also have that the derivative along $x$ is much lower than the derivative along $y$

$$kf \ll \frac{\partial f}{\partial y}, \qquad k \to 0$$

recalling $\partial_y f = f'$ we approximate

$$v'' \simeq \frac{v}{\delta^2}, \qquad b'' \simeq \frac{b}{\delta^2}$$

we use this in the equations

$$B_{0x} b'' \sim B_{0x} \frac{b}{\delta^2}, \qquad b B_{0x}'' \sim b \frac{B_{0x}}{a^2}$$

with $a$ spatial scale of the variation of $b$.
So

$$B_{0x} b'' \ll B_{0x}'' b \qquad (\delta \ll a)$$

**Equation of momentum inside**
Recall

$$\begin{cases} \gamma(v'' - k^2 v) = ik[B_{0x}(b'' - k^2 b) - b_y B_{0x}''] \\ \gamma b = B_{0x} ikv + \eta(-k^2 b + b'') \end{cases}$$

We consider only the small $k$ and $B_{0x} \sim 0$.
From the **momentum equation**

$$\gamma v'' - \gamma k^2 v = ik\left[B_{0x}b'' - k^2 B_{0x}b - bB''_{0x}\right]$$

we approximate it as (recall $kf \ll \partial f/\partial y$)

$$\gamma v'' \simeq ik(B_{0x}b'')$$

The **magnetic field equation** becomes

$$\gamma b \simeq ikB_{0x}v + \eta(b'' - k^2 b)$$

which gets approximated to

$$\gamma b \simeq \eta b''$$

# Outside

We assume

$$\eta \simeq 0, \qquad B_{0x} \gg v, b$$

**Equation of momentum** outside

$$\gamma(v'' - k^2 v) = ik\left[B_{0x}(b'' - k^2 b) - bB''_{0x}\right]$$

so we approximate ($v$ is of the same order of $b$, but the $b$ is multiplied by $B_{0x}$ which is huge)

$$ik\left[B_{0x}(b'' - k^2 b) - bB''_{0x}\right] \simeq 0$$

and dividing by $B_{0x}$ we get

$$b'' - k^2 b - \frac{B''_{0x}}{B_{0x}}b \simeq 0$$

The **magnetic field equation** is

$$\gamma b = ikB_{0x}v + \eta\left(b'' - k^2 b\right)$$

wich gets approximated to

$$\gamma b \simeq ikB_{0x}v$$

# Results

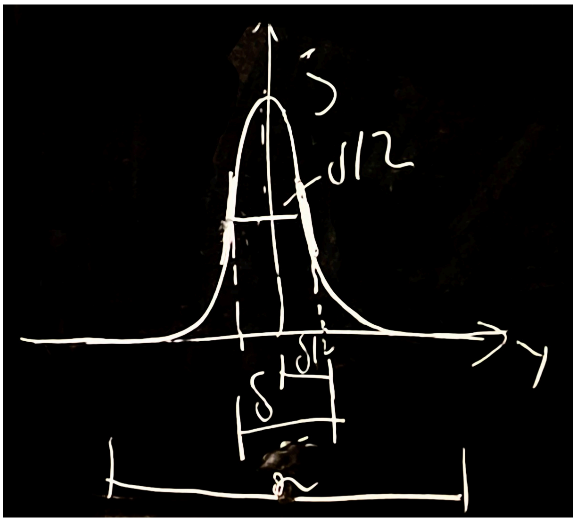
So inside we have

$$\begin{cases} \gamma v'' \simeq ik(B_{0x}b'') & (1) \\ \gamma b \simeq \eta b'' & (2) \end{cases}$$

and outside we have

$$\begin{cases} b'' - k^2 b - \dfrac{B''_{0x}}{B_{0x}}b \simeq 0 & (3) \\ \gamma b \simeq ikB_{0x}v & (4) \end{cases}$$

we remember that $B_{0x} = B_{0x}(y)$ so this makes things difficult.
PIC delta/2




We introduce a disconinuity parameter $\Delta'$

$$\Delta' \equiv a\frac{b'\left(\frac{\delta}{2}\right) - b'\left(-\frac{\delta}{2}\right)}{b\left(\frac{\delta}{2}\right)} = 2\frac{a}{b\left(\frac{\delta}{2}\right)}\left|b'\left(\frac{\delta}{2}\right)\right|$$

from symmetry.
We can write the second derivative of $b$ as
(we have $\delta$ with power $1$ because we are evaluationg it from the first derivative)

$$b''\left(\frac{\delta}{2}\right) \simeq \frac{b'\left(\frac{\delta}{2}\right) - b'\left(-\frac{\delta}{2}\right)}{\delta} = \frac{\Delta' \cdot b\left(\frac{\delta}{2}\right)}{a\delta}$$

and for $v$ we have

$$v'' \simeq -\frac{v\left(\frac{\delta}{2}\right)}{\delta^2}$$

it must be odd otherwise we have compressibility
The **moment equation inside** (eq 1) becomes (substituting $b''$ and v")

$$\gamma v'' \simeq ik(B_{0x}b'')$$

$$\implies -\frac{\gamma v\left(\frac{\delta}{2}\right)}{\delta^2} \simeq ikB_{0x}\left(\frac{\delta}{2}\right) \cdot \frac{\Delta' \cdot b\left(\frac{\delta}{2}\right)}{a\delta}$$

The **magnetic field equation outside** (eq 4) becomes

$$\gamma b \simeq ikB_{0x}v$$

$$\implies v \simeq \frac{\gamma b}{ikB_{0x}} = -\frac{i\gamma b}{kB_{0x}}$$

We need to match the solution inside and outside at $\delta/2$, so we substitute (eq 4) into (eq 1)

$$-\frac{\gamma}{\delta^{\cancel{2}}}\frac{\cancel{i}\,\gamma\, \cancel{b\left(\frac{\delta}{2}\right)}}{kB_{0x}\left(\frac{\delta}{2}\right)} = \cancel{i}\,kB_{0x}\left(\frac{\delta}{2}\right)\frac{\cancel{b\left(\frac{\delta}{2}\right)}}{a\cancel{\delta}}\Delta'$$

$$\implies \gamma^2 \simeq \left[kB_{0x}^2\left(\frac{\delta}{2}\right)\right]^2\left(\frac{\delta}{a}\right)\Delta'$$

Using Taylor we get (for $B_{0x}$)

$$B_{0x}\left(\frac{\delta}{2}\right) \simeq \cancel{B_{0x}(0)} + \frac{\delta}{2}B'_{0x}(0)$$

and substituting (don't confuse $\delta/2$ with $\delta/a$) (recall $\bar{B} = B_{0x}(y)\hat{x} + \bar{b}_1$)

$$\gamma^2 \simeq \left[k\left(\frac{\delta}{2}\right)B'^2_{0x}\right]^2\left(\frac{\delta}{a}\right)\Delta' =$$

$$= k^2B_0^2\left(\frac{aB'_{0x}}{2B_{0x}}\right)^2\left(\frac{\delta}{a}\right)^3\Delta' =$$

$$= k^2B_0^2A\left(\frac{\delta}{a}\right)^3\Delta'$$

with ($B_{0x} \to +\infty$)

$$A \equiv \left[\frac{aB'_{0x}}{2B_{0x}}\right]^2$$

$\gamma > 0$ and $\gamma \in \mathbb{R}$.
From the magnetic field equation inside (eq 2) we isolate the term $\delta/a$ as follow

$$\gamma b \simeq \eta b'' \qquad \implies \qquad \gamma \cancel{b} \simeq \eta\frac{\Delta' \cancel{b}}{a\delta}$$

$$\implies \gamma = \frac{\eta\Delta'}{a\delta} \qquad \implies \qquad \frac{\delta}{a} = \frac{\eta\Delta'}{a^2\gamma}$$

we substitute and we get

$$\gamma^2 \simeq k^2B_0^2A\left(\frac{\eta\Delta'}{a^2\gamma}\right)^3\Delta' \qquad \implies \qquad \gamma \simeq k^{2/5}B_0^{2/5}A^{1/5}\frac{\eta^{3/5}}{a^{6/5}}\Delta'^{4/5}$$

so

$$\gamma \sim k^{2/5}S^{-3/5}\Delta'^{4/5}$$

with the Lindquist number

$$S = \frac{aB_0}{\eta}$$

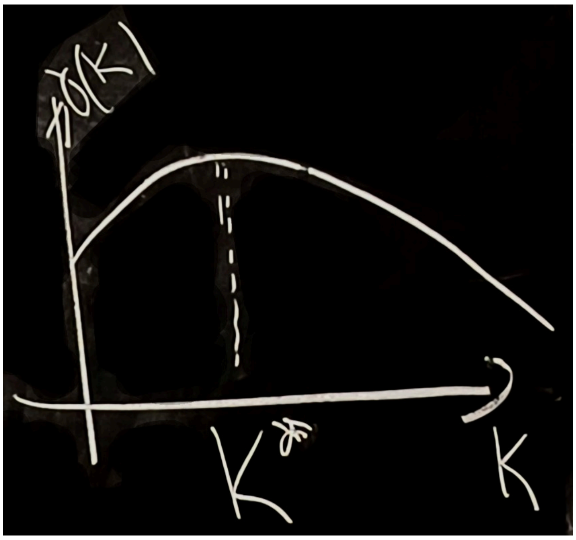
$\Delta'$ is a parameter which depends on $k$

$$\gamma \simeq S^{-3/5}f(k)$$

$f(k)$ is found numerically
PIC log gamma for fixed k


Bigger Lindquist number, lower growth rate

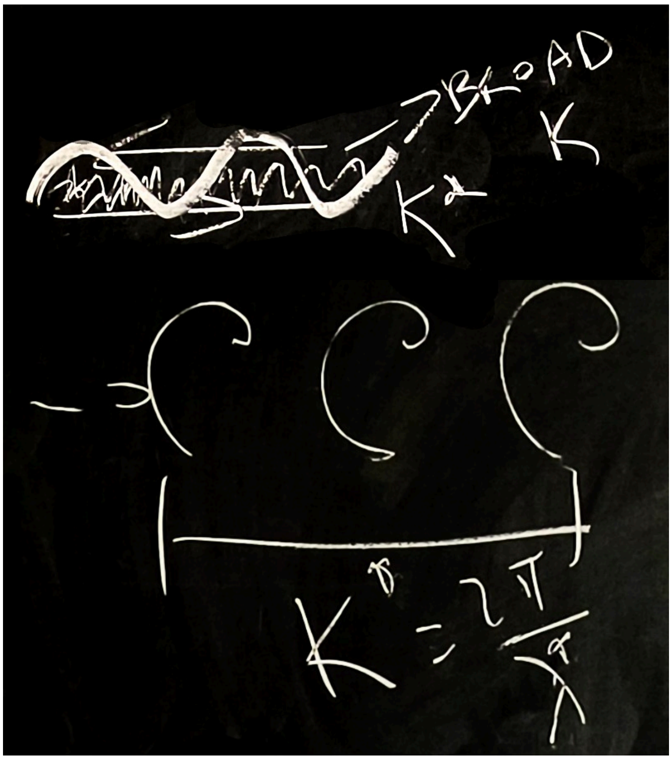
PIC gamma (k) at fixed S




k too low or too high do not work

there is a maximum growing mode $k^*$, which is a resonant $k$

PIC example

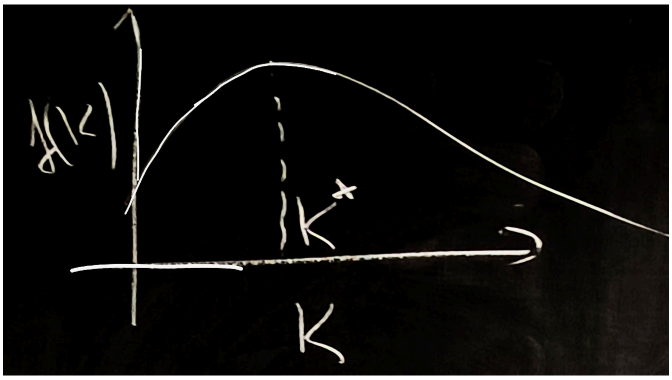


Now we will see nonlinearity
and finally a classic from plasma physics

We have two main results
Linear theory

$$\gamma \simeq S^{-3/5} f(k)$$

PIC k*




We used the decomposition

$$\bar{b}_1(x, y, t) = \bar{b}(y) e^{\gamma t - ikx}$$

$$\bar{v}_1(x, y, t) = \bar{v}(y) e^{\gamma t - ikx}$$

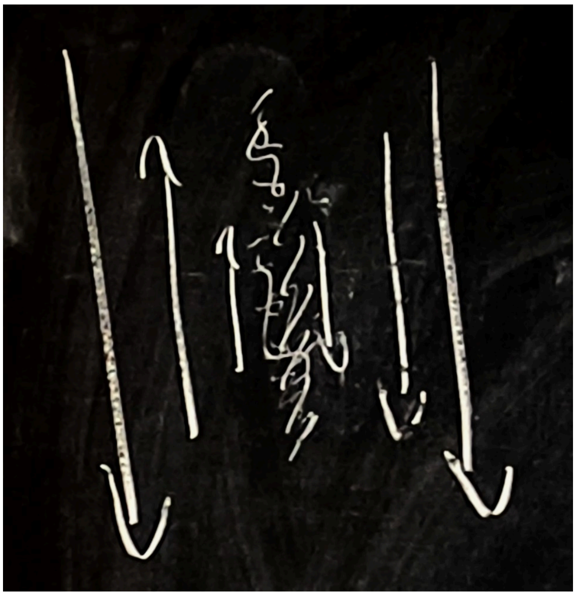
the time of growth is

$$\propto \frac{1}{\gamma}$$

Total magnetic field is

$$\bar{B} = B_{0x}(y)\hat{x} + \bar{b}_1$$

All our calculations are valid in the first quarter second (linear case).
It selects the maximal growing mode.
Lets consider a laminar magnetic field
PIC shear magnetic field with noise



of course we have $\lambda^* = 2\pi/k^*$

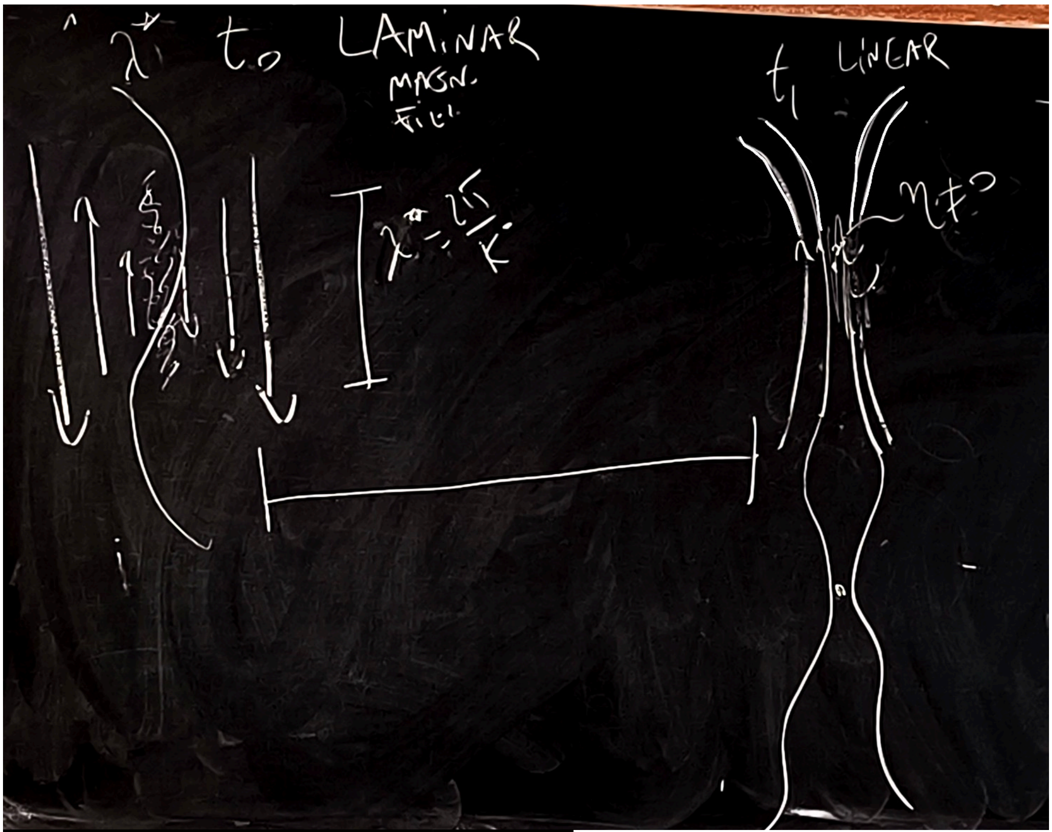
the field line will bend due to the $\lambda^*$ (we sum $B$ and $\lambda$)

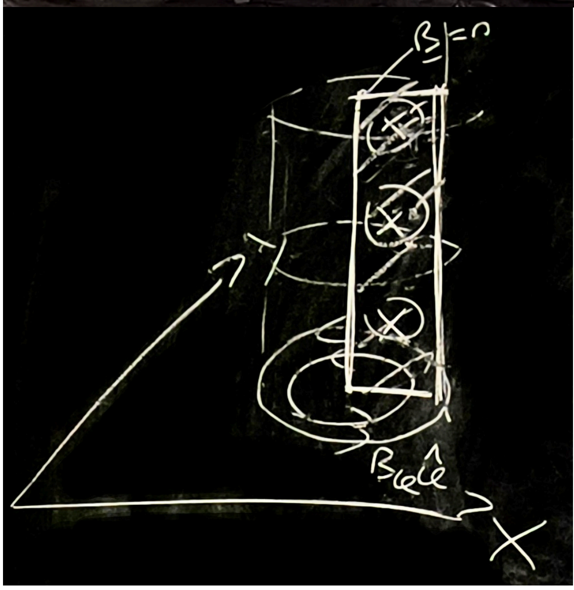
at $t^*$ this becomes ccompletely nonlinear

X means explosion

field lines break due to resistivity and they reconnect, the mean field is changing

Nonlinear magnetic reconnection




Notes on 2D magnetic reconnection

$$\bar{B} = \nabla \Psi \times \hat{z}$$

Poloidal magnetic flux, the magnetic potential is

$$\Psi = a_z$$

$\Psi$ measures how much magnetic field enters in the drawn plane of the cylinder (poloidal component)

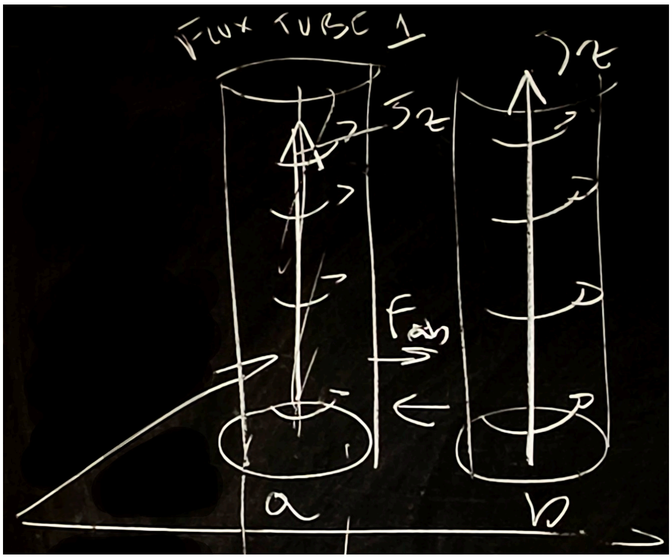
if $\eta = 0$, $\bar{U} = 0$

$$\frac{\partial \Psi}{\partial t} = -\cancel{\bar{U} \cdot \nabla \Psi} + \cancel{\eta \nabla^2 \Psi} = 0$$

Giovannelli asks

what happens with 2 coca cola cans?

PIC 2 cola cans




rotating flux tube has an inside current and

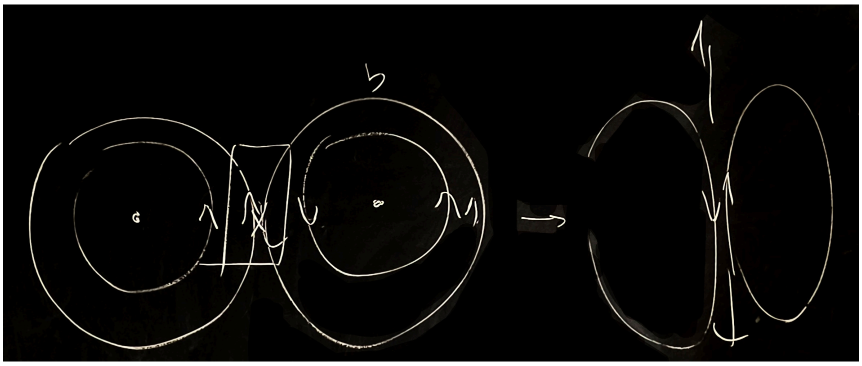
$$J_z = -\nabla^2 \Psi$$

two currents in the same direction (as in two cables w same current direction)

$$\bar{F}_{ab} = -\bar{F}_{ba} = J\bar{\ell} \times \bar{B}$$

and nothing depends on $\hat{z}$

PIC vew from the top



we will study the rectangle in a steady state because the flux tube are coming together slowly nonlinear solution.

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